Co-AI BEGINNER WORKSHOP · AGENTIC AI FUNDAMENTALS
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Build a Literature Research Agent using Claude AI

Pre-run set up before the session:

Sign in at claude.ai and confirm it works.

1. Go to **Settings → Connectors** (on some accounts this sits under **Customize → Connectors**).
2. Find **PubMed** in the connector directory and enable it. It needs no password, no API key, and no NCBI account.
3. Start a new chat and type: `Search PubMed for recent trials on semaglutide and cardiovascular outcomes.`
4. If you see results with PMIDs, you are ready. If not, message the organiser **before** the session.

Most connectors require a paid plan (Pro, Max, Team, or Enterprise). If you are on a free account, tell the organiser now so you can be paired with someone.

Part 1 — What an "agent" actually is

Strip away the hype and an agent is four things:

Component	In our build
A model	Claude, doing the reasoning
Tools	the PubMed connector — how it reaches real data
Instructions	your Project instructions — the standing method it follows
A verification step	the citation rule that makes output checkable

The fourth one is what separates a research agent from a chatbot. A model with no tools can only produce text that *looks* like a citation. A model with PubMed can produce one you can click.

Choosing a model

You are not writing code, so you pick from the model dropdown rather than per-step. Use **Claude Sonnet 5.5** — the best balance of speed and intelligence, and fast enough that a room full of people running searches doesn't stall.

The tier logic, so it transfers when you later build with the API:

Model	Cost (per million tokens)	Use it for
Claude Haiku 5.5	from $0.10 in / $0.50 out	high-volume classification, extraction, routing
Claude Sonnet 5.5	$2 in / $10 out	the everyday workhorse — **our choice**
Claude Opus 5.5	$4 in / $20 out	long-running agentic work
Claude Fable 5.1	$10 in / $50 out	demanding reasoning, long-horizon tasks

All four carry a 1M token context window. Reaching for the most expensive model is usually the wrong instinct: screening 40 abstracts is a sorting job, and Haiku does sorting at a fraction of the cost.

Part 2 — Create the Project

1. In the left sidebar, click **Projects → Create Project**. Name it `Literature Research Agent (based on you wish)`.
2. Open **Set project instructions** (sometimes **Add instructions**).
3. Paste the block below, verbatim.
4. Confirm PubMed is enabled for the Project.

Project instructions — paste this

```
You are a rigorous biomedical literature research assistant. Follow these
phases in order. Never skip screening. Never invent a citation, finding,
statistic, or sample size.

SCOPE FIRST
Before searching, confirm with me in one short round: the precise research
question; the date range (default: last 5 years); the study types that matter
(e.g. RCTs, systematic reviews, preclinical); and how many papers to include
(default: screen ~40 down to ~15).

SEARCH
Translate the question into 2-4 PubMed queries. Join concepts with AND and
synonyms with OR. Prefer MeSH terms for core concepts. Apply date and
publication-type filters as scoped.
Always report the total_count so I know the size of the field, and show me th
e
query_translation so I can see what PubMed actually searched.

SCREEN
Pull metadata for the candidate pool. Screen on title and abstract against th
e
research question. Drop off-topic or wrong-population papers and tell me
briefly why any notable one was dropped. Rank systematic reviews and large
```

```
trials above primary studies, and those above preclinical work.
For the 2-3 most central papers, retrieve the full text if a PMC ID exists, s
o
that claims about methods and limitations rest on more than the abstract.

SYNTHESISE
Give me three things:
1. A summary table — one row per paper: first author (year) | design | popula
tion
   and n | intervention | key finding | stated limitations | DOI link.
2. A narrative synthesis in prose, organised by theme rather than paper by
   paper. Lead with what the evidence collectively shows. Surface
   contradictions between studies instead of smoothing them over. Name the
   gaps.
3. An annotated bibliography — full citation plus 2-4 sentences on what each
   study did and why it matters.

GAP ANALYSIS
Do not give me a vague closing line about "more research being needed." Work
through these gap types explicitly and report only the ones the evidence
actually supports, with the paper or absence that justifies each:

- Knowledge gap: a question nobody appears to have asked. Say what you search
ed
  and how few results came back.
- Population gap: who is missing. Check age, sex, ethnicity, pregnancy,
  comorbidity, and disease severity across the included studies.
- Methodological gap: small samples, no randomisation, short follow-up,
  surrogate rather than clinical endpoints, no active comparator, unblinded
  outcome assessment.
- Measurement gap: studies measuring the same construct in incompatible ways,
  so results cannot be pooled.
- Contradiction gap: studies that genuinely disagree. Name both sides and say
  what might explain the difference.
- Mechanistic gap: an effect that replicates but is unexplained.
- Temporal gap: evidence predating a change in standard of care, or a field
  where the newest work is several years old.
- Setting gap: all evidence from high-income or single-country settings.
- Translation gap: efficacy shown under trial conditions with no
  effectiveness, implementation, or cost data.
- Reporting-bias signal: outcomes concentrated in industry-funded work, or
  registered trials with no published results.

Then convert the three most important gaps into concrete, runnable PubMed
queries I could execute today, and say what each would tell me.

CITATION RULE — NON-NEGOTIABLE
Attribute PubMed explicitly whenever you use its data. Give every referenced
article a clickable DOI link (https://doi.org/...), or its PubMed page
```

```
(https://pubmed.ncbi.nlm.nih.gov/<pmid>/) when no DOI exists.
If an abstract does not state something, write "not reported in abstract"
rather than filling the gap.

BEFORE YOU FINISH
Self-check: does every cited paper have a working link? Is every claim tracea
ble to a retrieved article? Are contradictions visible? Is the search strateg
y stated clearly enough that I could reproduce it?
```

Teaching point: the instructions are the agent. Everything that makes this trustworthy is a sentence someone chose to write. Participants should expect to edit this block for their own field.

Part 3 — Run it

Start a chat inside the Project and paste one question. Pick from these, or bring your own:

- `What does the evidence say about GLP-1 receptor agonists and kidney outcomes in type 2 diabetes? Last 5 years, trials and systematic reviews only.`
- `Summarise recent evidence on faecal microbiota transplant for recurrent C. difficile infection.`
- `What is known about tau-targeting therapies in Alzheimer's disease? Focus on clinical trials.`

Let the agent ask its scoping questions and answer them. Watch what it does — the tool calls are visible, and that visibility is the point.

It will take 5-7 minutes to complete the run and provide you with the results

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Part 4 — Verifying authenticity

This is the part that matters most, and the reason the workshop exists. An AI that fabricates a plausible citation is more dangerous than one that admits it found nothing.

Check 1 — Read the query translation

PubMed rewrites your keywords. Asking for `semaglutide cardiovascular outcomes` actually searches an expansion across `semaglutide[Supplementary Concept]`, `semaglutide[MeSH Terms]`, `cardiovascular system[MeSH Terms]` and several free-text variants. Read that translation. If it expanded into something you didn't intend, your results are answering a different question than you asked.

Check 2 — Click the links

Every PMID resolves to `https://pubmed.ncbi.nlm.nih.gov/<pmid>/` and every DOI to `https://doi.org/<doi>`. Pick two citations at random and click them. Title, journal, and year must match what the agent told you. Thirty seconds, every time.

Check 3 — Reverse-verify a reference (the one to actually demo)

Hand the agent a citation and ask it to confirm the paper exists:

```
Here is a reference from a draft manuscript. Use PubMed's citation lookup to
confirm it exists and return the PMID and DOI. If you cannot match it, say so
plainly — do not guess.
```

Then try it with a reference you have invented: a real-sounding author, a real journal, a plausible year. A careful agent comes back empty-handed. **That empty result is the correct answer**, and recognising it is the skill being taught.

This also happens to be a genuinely useful tool. Run your own reference lists through it.

Check 4 — Before you reuse anything

Ask the agent to check copyright and licensing status before you quote or reproduce figures. Open-access licences such as CC BY 4.0 permit reuse; many articles do not.

What this does *not* protect you from

Verification confirms a paper **exists and was described accurately**. It says nothing about whether the study was any good, whether the field has moved on, or whether a retraction has been issued. PubMed indexes retracted papers. The agent compresses the literature; it does not appraise it.

Data handling — read before pasting anything

Claude.ai is a cloud service. Under our organisation's policy you may use it with data classified up to **ARES Private**. Cowork permits up to **ARES Confidential**.

For this workshop that is a non-issue: published literature is public. But do not paste unpublished manuscripts, grant applications, participant data, or anything above ARES Private into a Claude.ai chat. Searching *for* papers is fine. Uploading your confidential draft is not.

Troubleshooting

Symptom	Fix
PubMed missing from the directory	Check your plan; most connectors need Pro, Max, Team, or Enterprise

Symptom	Fix
Connector not available inside the Project	Check the Project is private — connectors don't work in shared or public project chats
Searches return almost nothing	Query too narrow. Drop a filter, add OR synonyms, widen the dates
Tens of thousands of hits	Add MeSH terms and a publication-type filter
"Full text not available"	Only a subset of PubMed has a PMC ID. Abstract-only is normal
Agent stops citing DOIs	Instructions drifted out of context in a long chat. Start a fresh chat in the Project

Where to go next

- Rewrite the Project instructions for your own field and study types
- Add a screening step with explicit inclusion and exclusion criteria
- Ask the agent for the search strategy as a reproducible methods paragraph
- When you outgrow no-code, the same four components — model, tools, instructions, verification — port directly to the API

Appendix A — Finding research gaps that are actually there

Take-home reference. We will not work through this in 30 minutes, but it is the part you will use most after the session.

An agent asked to "identify gaps" will produce fluent filler unless you tell it what a gap looks like. Worse, a gap is often an **absence**, and absences are exactly what a language model is worst at noticing — nothing in the retrieved text says "no one has studied this." You have to search for the hole deliberately.

Gap type and the question it asks	How to detect it with PubMed
Knowledge — has anyone asked this at all?	Run the narrow query and read `total_count`. A handful of hits in an otherwise busy field is the signal
Population — who was left out?	Compare MeSH terms across included papers. Missing `Aged`, `Female`, `Pregnancy`, `Child`, or comorbidity terms is itself the finding
Methodological — how good is the evidence, not how much?	Filter by publication type. If `Randomized Controlled Trial[Publication Type]` returns nothing, the field rests on observational work
Measurement — can these results even be combined?	Read the outcome definitions in the abstracts. Different instruments for one construct blocks meta-analysis
Contradiction — do studies disagree?	Ask the agent to list findings by direction of effect rather than paper by paper

Gap type and the question it asks	How to detect it with PubMed
Mechanistic — do we know *why*?	Search the intervention alongside mechanism terms. Clinical hits plus no preclinical hits is a real gap
Temporal — is this evidence stale?	Slice the same query by date range and compare counts. A field that stopped publishing years ago either resolved the question or abandoned it
Setting — where was this done?	Check affiliations and population descriptions. Add a geographic term and see what survives
Translation — does it work outside a trial?	Search the intervention alongside implementation, effectiveness, or cost terms
Reporting bias — what is missing from the record?	Note funding sources across the pool, then check ClinicalTrials.gov for registered trials with no publication

The detection trick worth teaching

Most gap types above are found by **running a search you expect to fail**. Prompt the agent this way:

```
Run a deliberately narrow search for [intervention] in [understudied
population]. Report total_count and the query_translation. If the count is
very low, do not pad the answer with adjacent literature — tell me the gap is
real and show me the query that demonstrates it.
```

That last sentence matters. Without it the agent helpfully substitutes nearby papers and the gap disappears.

Separating a real gap from a search failure

A thin result set has two very different causes, and confusing them produces false claims:

- **A genuine gap** — the query was sound, synonyms and MeSH terms were covered, and the literature is still thin.
- **A bad query** — a misspelling, a drug's generic name when the field uses the brand, an overly restrictive filter, or a MeSH term that doesn't exist.

Always check the `query_translation` before believing a gap. If PubMed failed to map your core concept to a MeSH term, you have found a vocabulary problem, not a research gap.

Appendix B — Turning a gap into your next search

A gap statement is only useful if it implies an action. Ask the agent to close the loop:

```
For each gap you identified, give me: the PubMed query I should run next, wha
t
result would confirm the gap is real, and what result would mean I was wrong.
```

Patterns that work:

Gap found	Next search
No trials in older adults	`[intervention] AND aged[MeSH Terms] AND clinical trial[Publication Type]`
Evidence looks outdated	Same query, `date_from` set to two years ago, sorted by publication date
Mechanism unclear	`[intervention] AND (mechanism OR signaling OR pathway)`
No real-world data	`[intervention] AND (effectiveness OR implementation OR "real world")`
Possible contradiction	Pull the two disagreeing papers, then use related-article lookup on both

Finish by asking for the search strategy written as a methods paragraph — databases, queries, date range, filters, counts, and screening decisions. That paragraph is what makes the work reproducible, and it is the thing reviewers ask for.